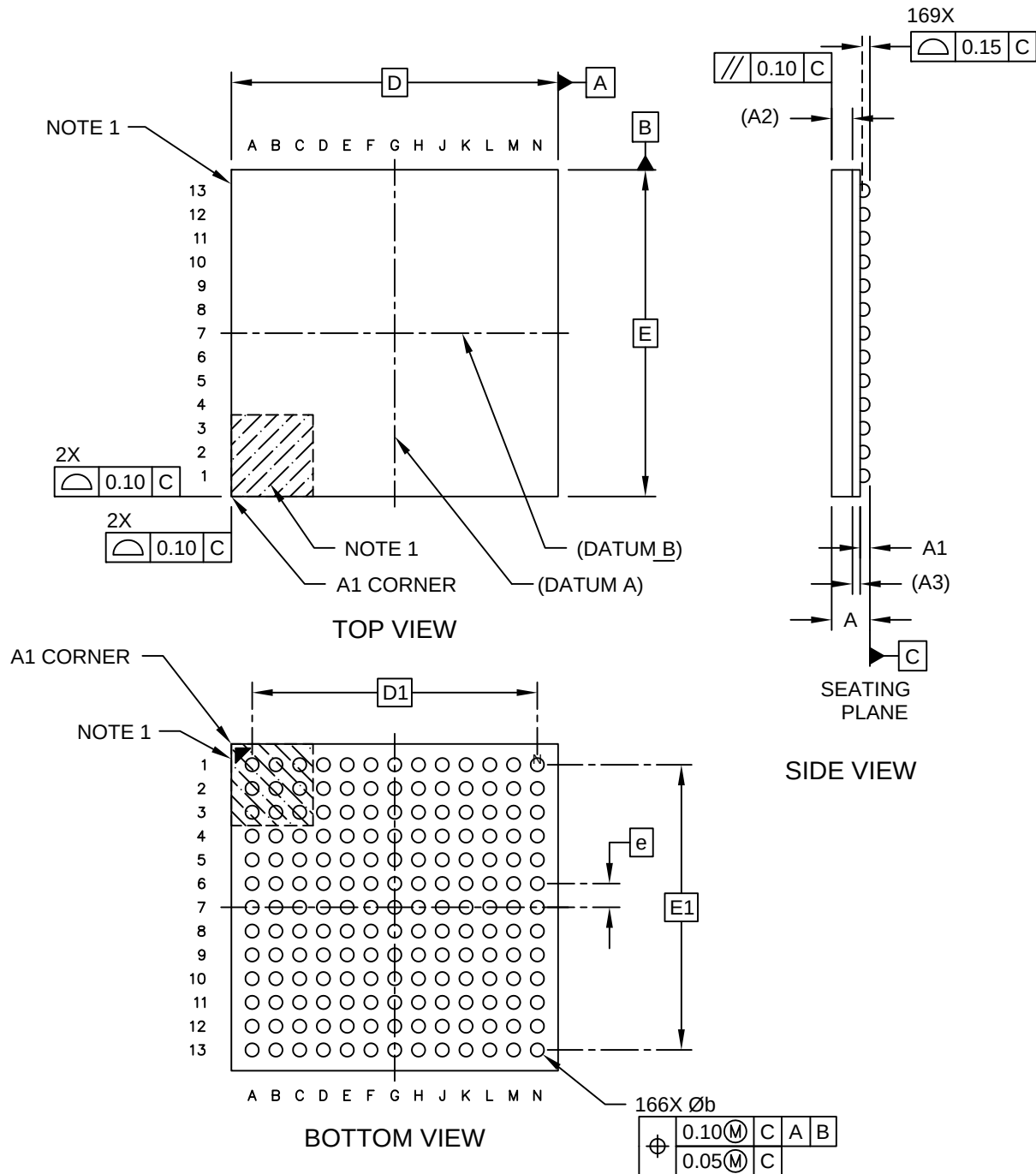


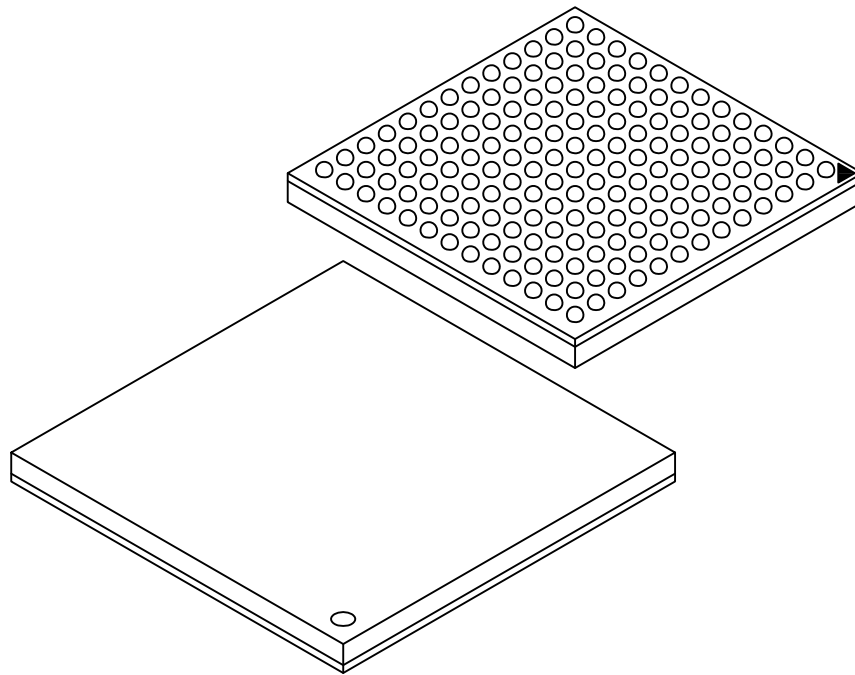
169-Ball Low Profile Fine Pitch Ball Grid Array (HF) - 11x11x1.4 mm Body [LFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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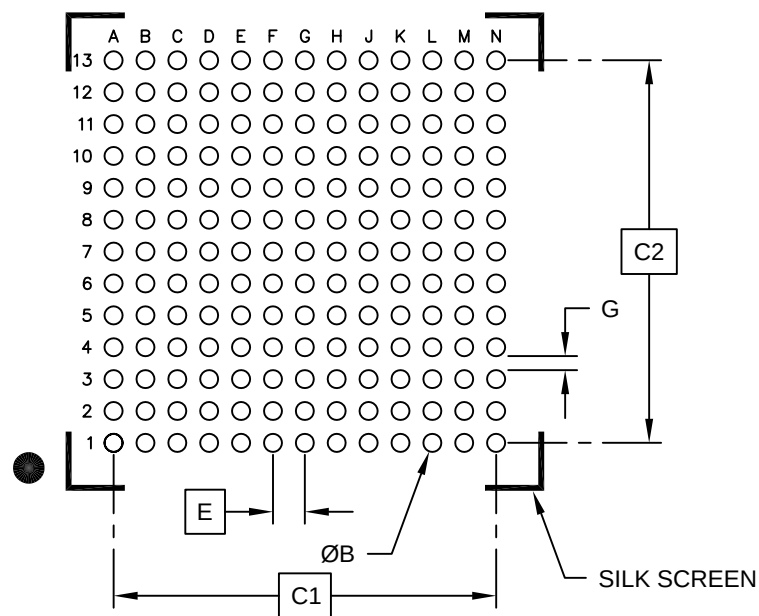
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals (Balls)	N	169		
Pitch	e	0.80 BSC		
Overall Height	A	1.17	1.285	1.40
Terminal (Ball) Height	A1	0.25	0.325	0.40
Mold Cap Thickness	(A2)	0.70 REF		
Substrate Thickness	(A3)	0.26 REF		
Overall Length	D	11.00 BSC		
Overall Width	E	11.00 BSC		
Overall Ball Pitch	D1	9.60		
Overall Ball Pitch	E1	9.60		
Ball Diameter	b	0.40	0.45	0.50

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

169 Ball Low Profile Fine Pitch Ball Grid Array (HF) - 11x11x1.4 mm Body [LFBGA]

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	9.60 BSC		
Contact Pad Spacing	C2	9.60 BSC		
Contact Pad Diameter (X169)	B	0.40	0.45	0.50
Pad-to-Pad Clearance	G	0.30		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.